Keystone Environmental
Knowledge-Driven Results

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Treatment of Wood Waste Landfill Leachate Using Engineered Wetlands

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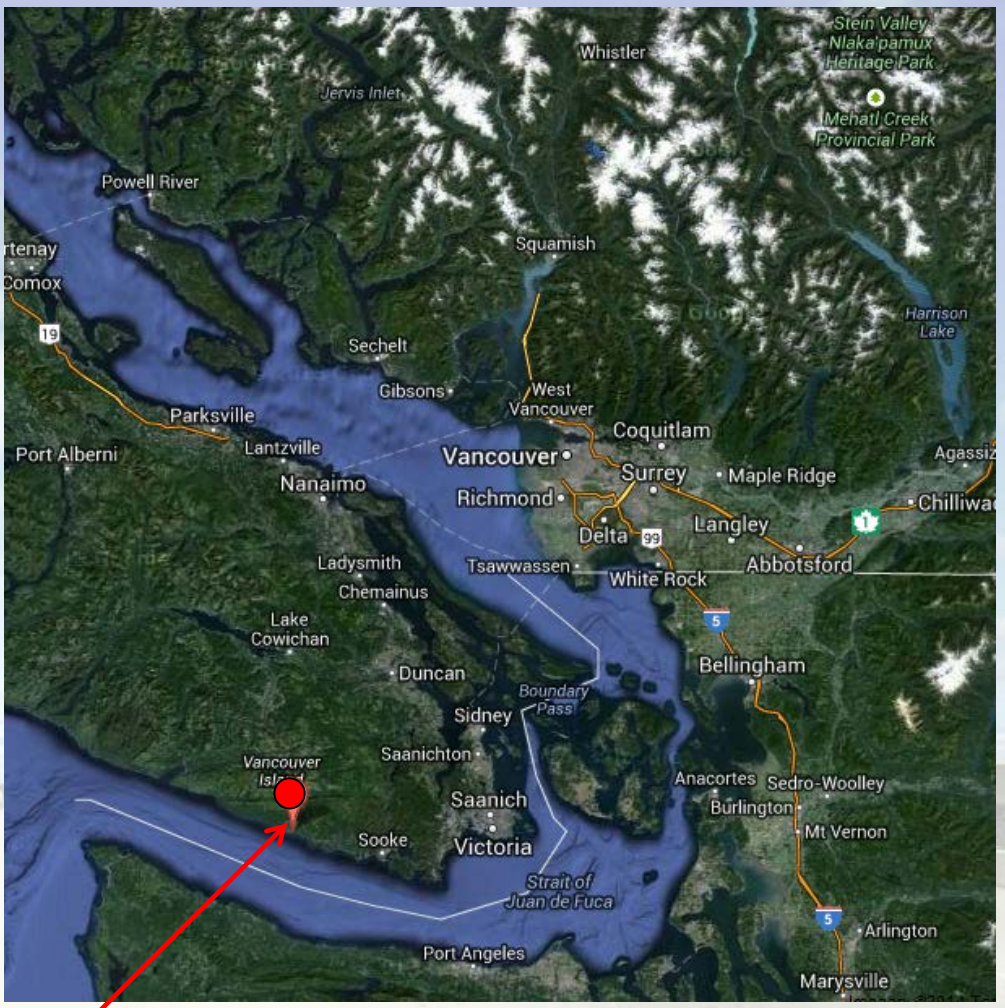
Site Description

- Former log sort and logging operations centre with on-site landfill.
- Located on the southwest coast of Vancouver Island, along the south shore of the Jordan River and fronting the Strait of Juan de Fuca.



Jordan River

Image from https://maps.google.ca

Wood Waste Landfill

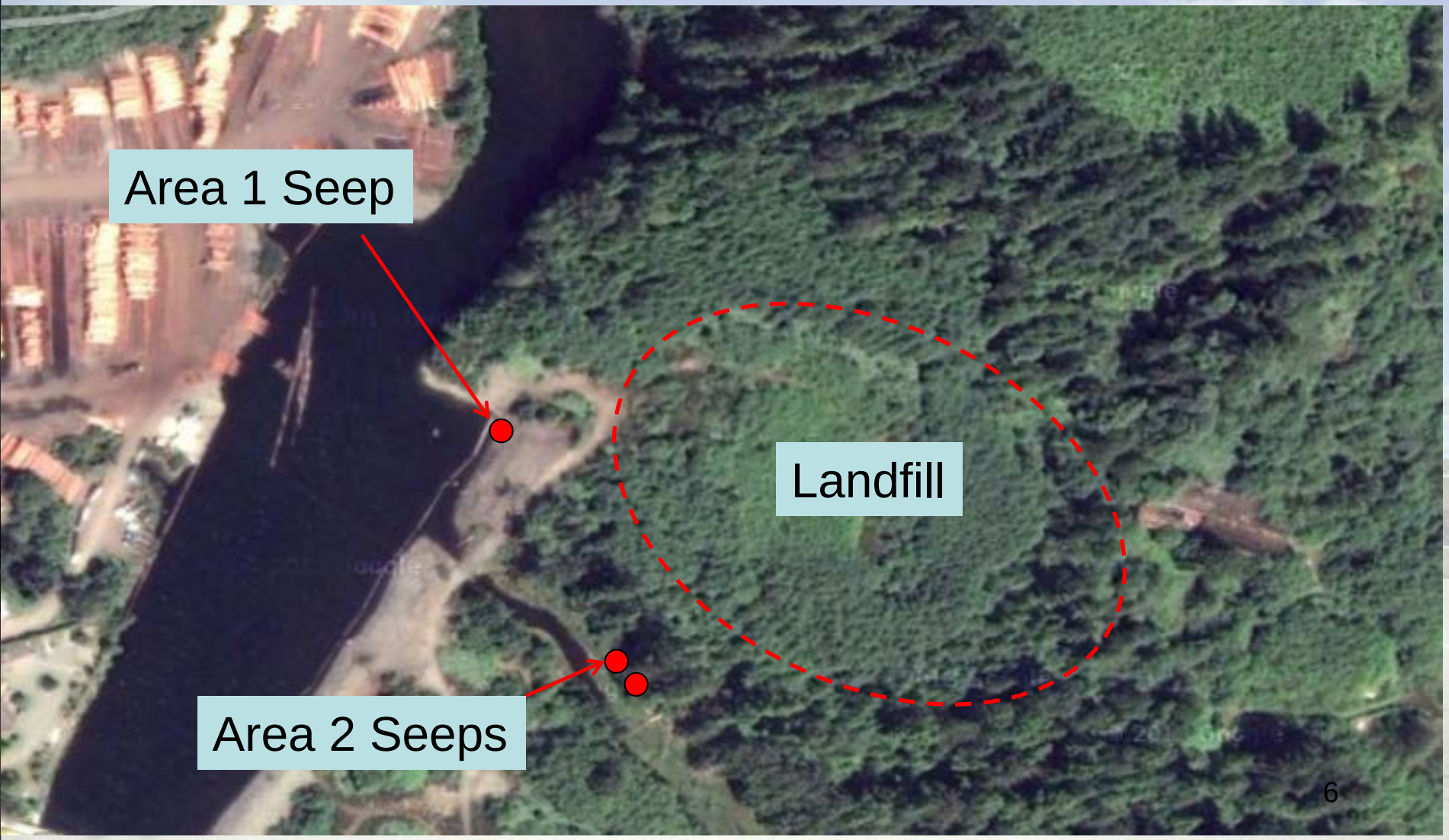
- A wood waste landfill is located on the property, adjacent to Jordan River to the west and a spillway to the south that was associated with a former hydroelectric generating station

Image from https://maps.google.ca

Remediation Drivers

- Upon closure, a portion of the property was to be sold to the Capital Regional District (CRD) for future park use.
- In the purchase sale agreement, the Seller agreed to complete remediation of the property and deliver a Certificate of Compliance.
- Investigations found that landfill leachate seeps were discharging to the spillway and to the Jordan River.
- The leachate contained dissolved sulphides as H_2S at concentrations up to 50 times the *British Columbia Water Quality Guidelines* (BCWQG) for freshwater and marine aquatic life environments

Seep Locations



Sulphides Concentrations at Landfill Seeps

Sample ID Lab Sample ID Date Sampled	RDL	Area 1 Seep 16-JUL-12	Area 2 Seep 1 16-JUL-12	Area 2 Seep 2 16-JUL-12	WQG AW_{FW}	WQG AW_{M}
Total Sulphide	2	154	311	49	n/g	n/g
Dissolved Sulphide	2	122	82	9.8	n/g	n/g
Dissolved Sulphide as H_2S*	-	**108**	**38**	**5**	2	2

Anions notes

NOTES:

All concentrations in micrograms per litre (µg/L)

Seep at Jordan River – Area 1

Seep into Spillway – Area 2

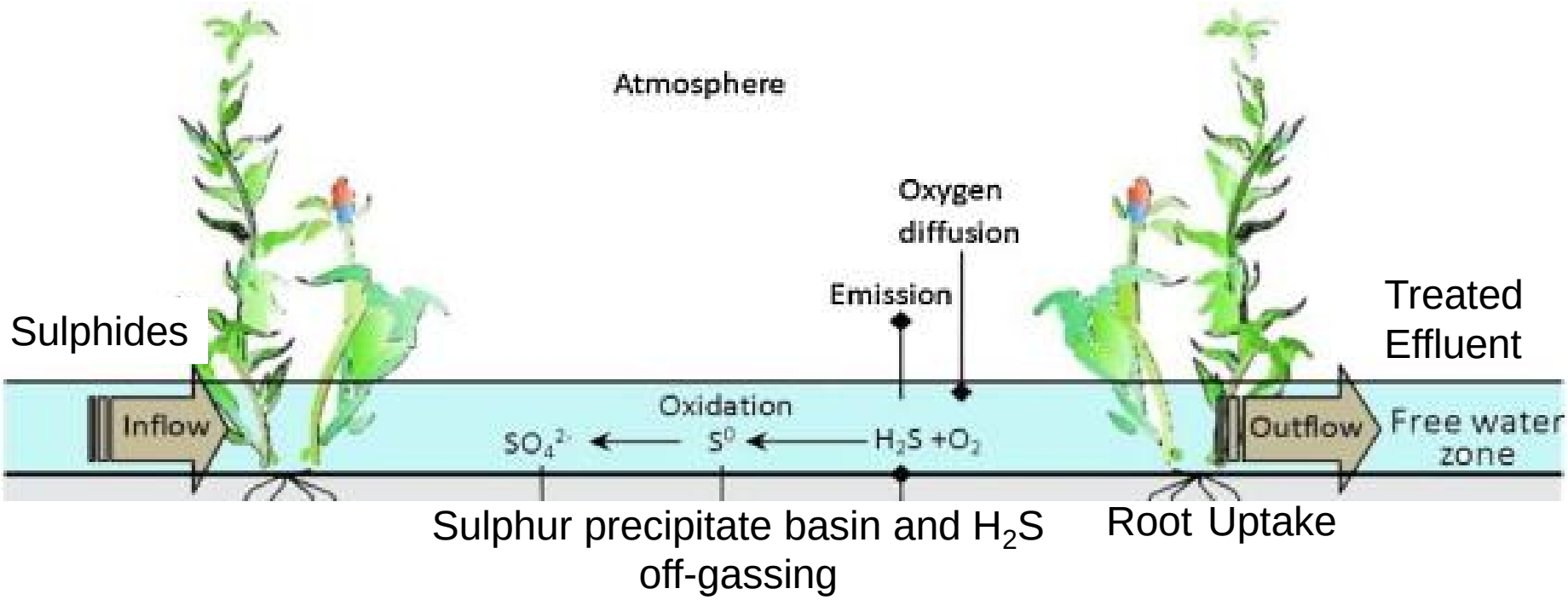
Remediation Challenges

- The leachate was expected to continue to be released from the landfill for several years following the land sale.
- Following completion of the sale, the Purchaser would not have staff operating from the Site.
- The remediation strategy needed to be <u>low maintenance</u> and <u>sustainable</u> over the long term.

Remediation Challenges

- Physical, chemical and biological treatment systems were considered
- Problem 1 = typically require ongoing operator attention
- Problem 2 = the Site is in a remote location for access
- Solution = Engineered Wetlands.
  - Minimal maintenance
  - No additional operator duties
  - No additional utility or chemical requirements

Typical Wetland Schematic



Three main processes in wetlands for sulphides removal:
1) sulphur precipitate; 2) H_2S off-gassing; and 3) root uptake

Shubiao Wu, Peter Kuschk, Arndt Wiessner, Jochen Müller, Rania A.B. Saad, Renjie Dong, Sulphur transformations in constructed wetlands for wastewater treatment: A review, Ecological Engineering, Volume 52, March 2013, Pages 278-289, ISSN 0925-8574, 10.1016/j.ecoleng.2012.11.003. (http://www.sciencedirect.com/science/article/pii/S0925857412003448)

Design Objectives

- Primary Objective: To reduce dissolved sulphides (as H_2S) concentrations to levels below the BCWQG.
- Secondary Objective: To enhance the existing environment by providing additional habitat for aquatic life.

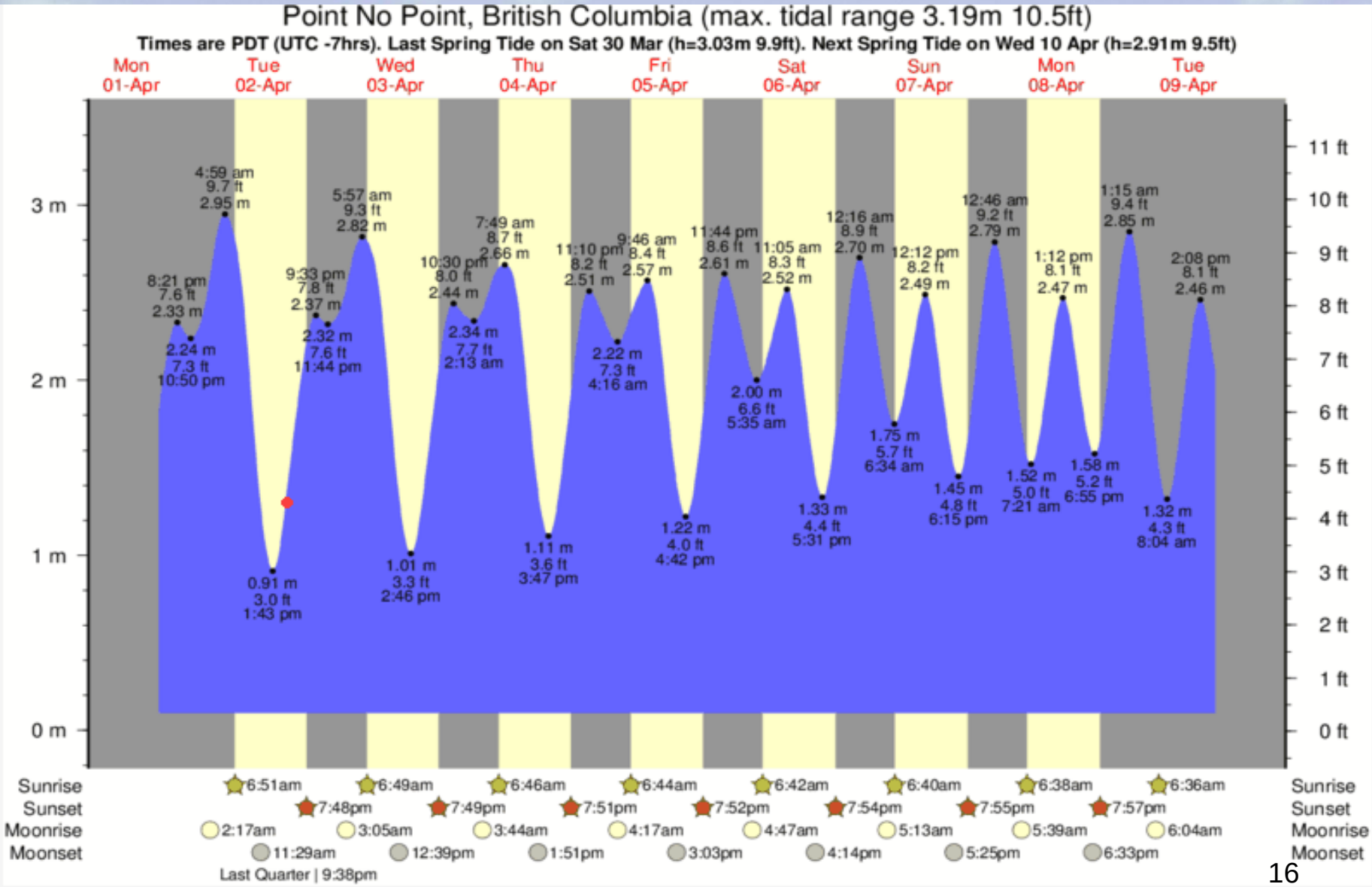
Design Considerations – Wetland Holding Capacity

- Calculating the minimum area required at the base of each wetland for adequate tidal holding capacity, and to maximize plant growth zones.

Design Considerations – Tidal Fluctuations

- Determining seasonal tidal fluctuations to make sure the wetlands have consistent influx and drainage.

Design – Tidal Fluctuation



http://www.surf-forecast.com/breaks/Jordan-River/tides/latest

Design Considerations – pH Neutralization

- Calculating the required holding time to allow for pH neutralization of the seep.
- Neutralizing the discharge water facilitates the chemical reaction for the transformation of dissolved sulphides to H_2S gas.

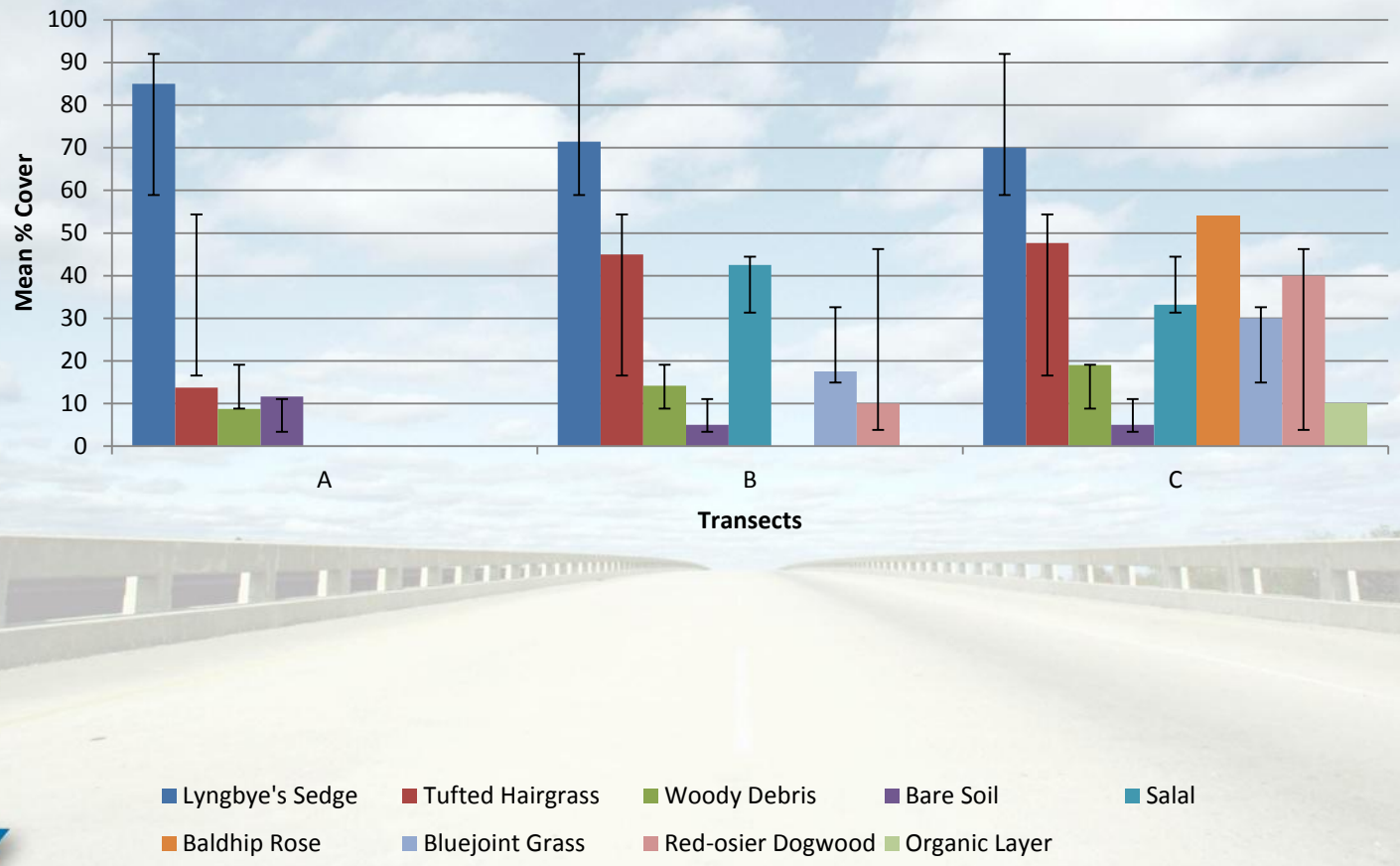
Design Considerations – H_2S Off-Gassing

- Increasing the seep discharge path to increase aeration for off-gassing the H_2S from the sulphides in solution.

Design Considerations – Vegetation

- Re-using native vegetation for the replanting of the engineered wetland to take advantage of the root oxygen release rates for further oxidation of the dissolved sulphides as H_2S.

Vegetation Transects to Determine Planting Plan



Design Calculations

Design Basis

	Influent Seep H_2S Concentration (µg/L)	Seep Flow Rate (L/min)	H2S Loading (µg/min)	Effluent Target H_2S Concentration (µg/L)
Area 1	108	2	323	2
Area 2	38	1.5	8.25	

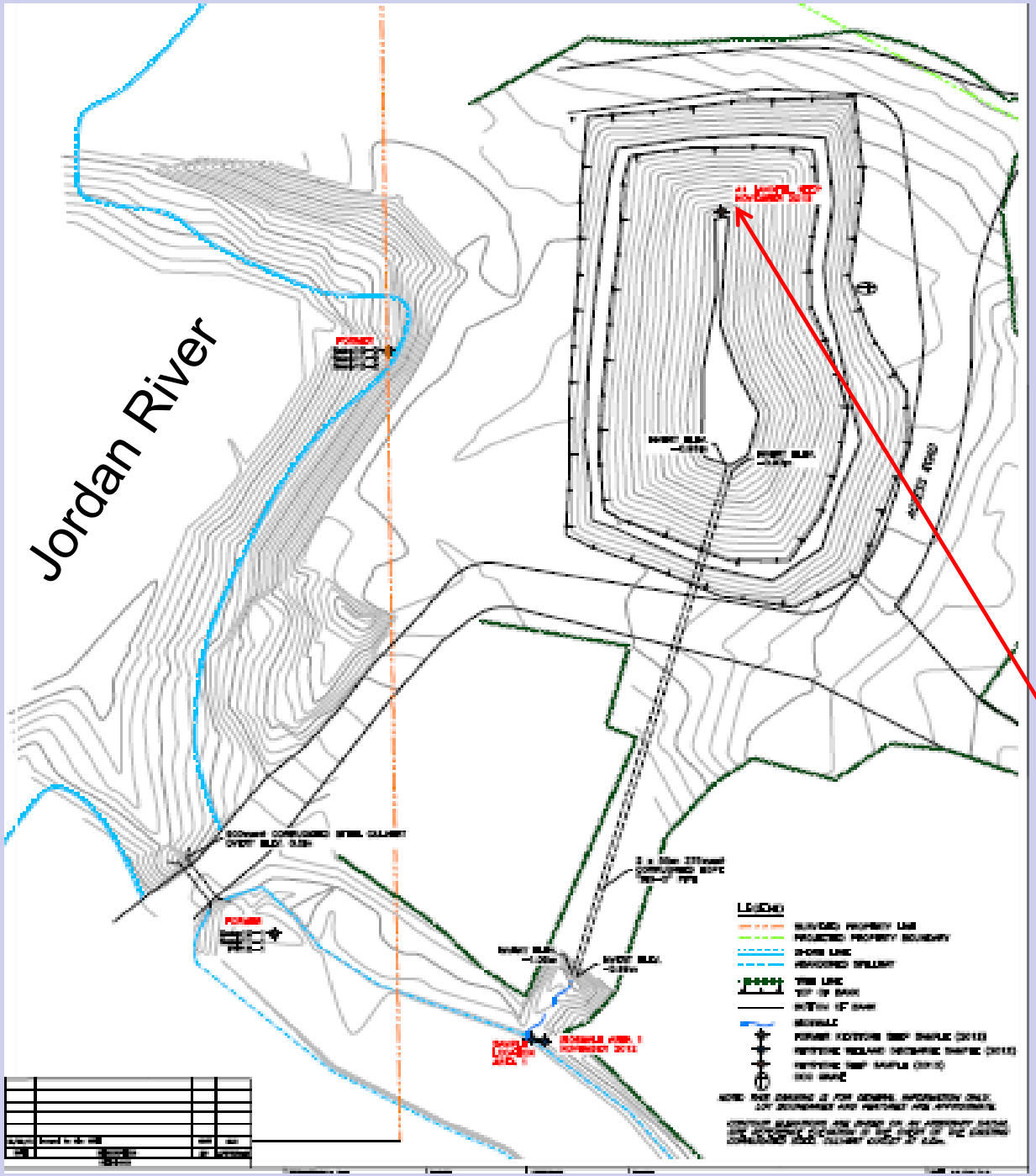
	Wetland Volume (m^3)	Wetland Area (m^2)	Depth (m)
Area 1	108	2	323
Area 2	38	1.5	8.25

Wetland volme and area calculations include a Safety Factor of 1.5

- Wetland area size based on settling velocity of sulphur compounds after aeration and pH neutralization

Engineered Wetlands

Engineered Wetland for Area 1



Intercepted Seep

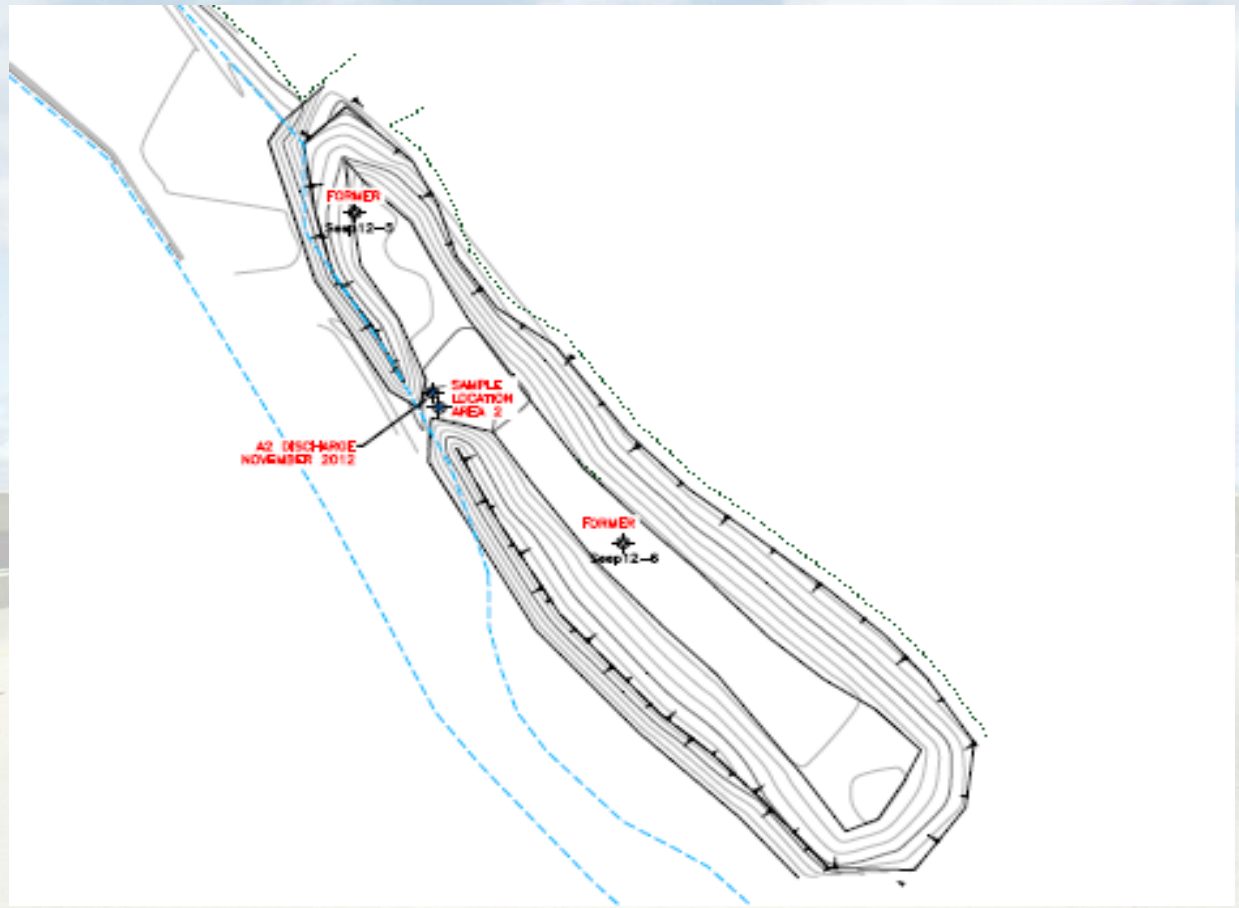
Wetland Area 1 – During Construction Before Planting

Wetland Area 1 – During Planting

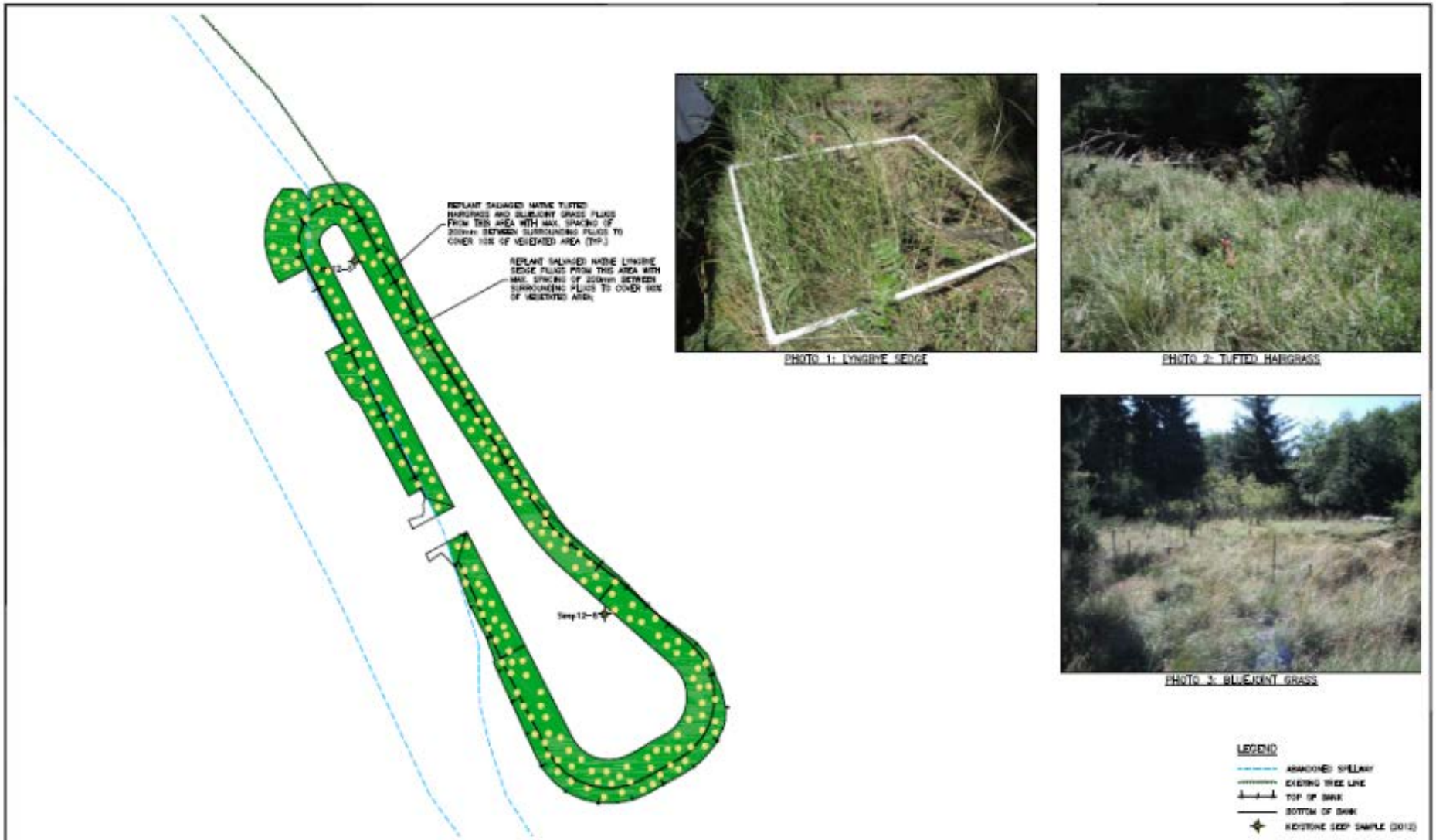
Wetland Area 1 – Installing the Culverts

Engineered Wetland for Area 2

Wetland Area 2



Wetland Area 2 – Planting Plan



Wetland Area 2 – Plant Survival

Tufted Hairgrass

Lyngbye's Sedge

Post Construction - Monitoring

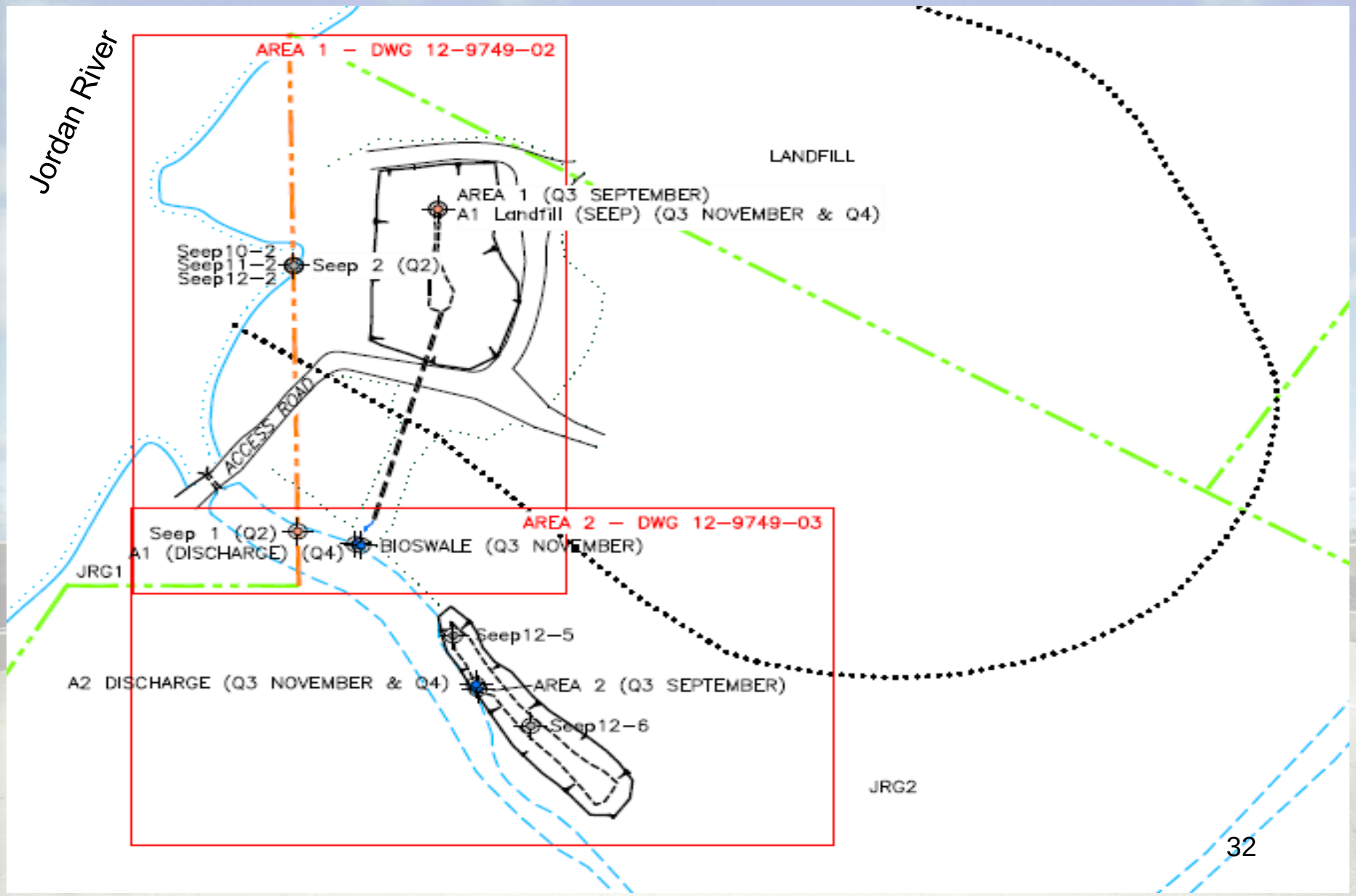
- The seep in Area 1 was intercepted and the flow is now visible into the north end of the wetland

Post Construction - Monitoring

- The seeps in Area 2 were intercepted and the flow is no longer visible

Areas of former seeps

Sampling Locations



Post Construction - Sampling

- Same as pre-construction sampling parameters
- Conducted field titrations for dissolved sulphides at the seep in Area 1, discharge of Area 1 and discharge of Area 2
- Dissolved sulphides as H_2S reduced from 54.6 ug/L at the seep to 1.9 ug/L at the discharge in Area 1

Post Construction – Sampling Results

Sample ID	Dissolved Sulphide (ug/L)	Dissolved Sulphides as H_2S (ug/L)	Target BC WQG AW_M at Discharge (ug/L)
AREA 1 SEEP	58	54.6	
AREA 1 DISCHARGE	2.8	1.9	2
AREA 2 DISCHARGE	<0.2	<0.0011	2

Engineered Wetlands - Results

- Concentrations of dissolved sulphides (as H_2S) in the wetlands discharge was less than the BCWQG during sampling in both wet and dry seasons.

Engineered Wetlands - Results

- Successful results were submitted to the Ministry of Environment in December 2012.
- A Certificate of Compliance was issued in 2013 and the property was transferred to the CRD.
- Future plans are to create park land for public use.

Thank You!

Questions?

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